Problem 0

Let

$$\mathbf{v} = \langle 1, \sqrt{3} \rangle, \qquad \mathbf{r}(t) = \left\langle \sin\left(\frac{\pi t}{6}\right), \cos\left(\frac{\pi t}{6}\right) \right\rangle.$$

(a) Solve the system of equations $\lambda \mathbf{r}(t) = \mathbf{v}$.

(b) Interpret your answer to part (a) geometrically.

Problem 1

In a simplified model, a planet at distance r from its star receives radiation at a rate per unit area

$$F(r) = \frac{L}{4\pi r^2}.$$

Its equilibrium temperature is

$$T(F, A) = \left(\frac{(1 - A)F}{4\sigma} \right)^{1/4},$$

where A is the fraction of incoming radiation the planet reflects, L is the star's luminosity, and σ is the Stefan–Boltzmann constant, with L and σ held constant.

(a) Compute $\dfrac{\partial T}{\partial A}$ and $\dfrac{\partial T}{\partial r}$.

(b) With the star at the origin, $r(x, y) = \sqrt{x^2 + y^2}$. For fixed A, find the direction of greatest temperature increase at $(0, 3)$.

(c) With A fixed, find the instantaneous rate of change of temperature at $(0, 3)$ if the planet has velocity $\langle 1, 2 \rangle$.

Problem 2

Let

$$f(x, y, z) = x^2 + y^2 + z^2 - 2x - 4y - 2z.$$

(a) Find the absolute maximum and minimum of f on the region $x \geq 0$, $y \geq 0$, $z \geq 0$, or explain why they do not exist.

(b) Repeat part (a) for the region $x, y, z \leq 0$.

Problem 3

A small workshop's daily output is modeled by

$$Q(x, y) = xy^2,$$

where x is the number of hours of labor and y is the number of hours of machine time. Labor costs \$20 per hour, and machine time costs \$10 per hour. Spending its entire daily budget of \$240 subjects the workshop to the constraint

$$20x + 10y = 240.$$

(a) Find the allocation of labor and machine time that maximizes daily output, and find that output.

(b) If the daily budget doubles, find the new optimal allocation and maximum output. Does the maximum output also double?